CBSE Class 12 Biology

Most Repeated Questions 2026 for Board Exam

Chapter 2 Sexual Reproduction in Flowering Plants

1. Angiosperms bearing unisexual flowers are said to be either monoecious or dioecious. Explain with the help of one example each.

2. Draw a diagram of a mature microspore of an angiosperm. Label its cellular components only.

OR

Draw a labelled diagram of a mature pollen grain.

3. A bilobed, dithecous anther has 100 microspore mother cells per microsporangium. How many male gametophytes this anther can produce?

4. A pollen grain in angiosperm at the time of dehiscence from an anther could be 2-celled or 3-celled. Explain, how are the cells placed within the pollen grain when shed at a 2-celled stage.

5. Describe the process of megasporogenesis up to fully developed embryo sac formation in an angiosperm. **(All India 2019)**

OR

Where does the process of megasporogenesis start in an angiosperm? Describe the process up to the formation of the embryo sac.

6. (i) Describe the sequence of the process of microsporogenesis in angiosperms.

(ii) Draw a labelled diagram of a 2-celled final structure formed.

OR

Trace pollen grain development from sporogenous tissue in the anther.

7. List the different types of pollination depending upon the source of pollen grain.

OR

Differentiate between autogamy, geitonogamy and xenogamy.

8. (i) Draw an LS of the pistil showing the pollen tube entering into the embryo sac. Label the following. **(All India 2019)**

(a) Nucellus

(b) Antipodals

(c) Synergids

(d) Micropyle

(ii) Write the functions of the following

(a) Synergids

(b) Micropyle

9. A biology student Ashu, read an article on apples being a false fruit. He asked his teacher about how a fruit can be called false and was explained about the development of fruits.

(i) What are false fruits? Give an example.

(ii) What is meant by parthenocarpy? Name a fruit developed by this method.

(iii) What are the values shown by Ashu?

Chapter 3 Human Reproduction

1. Why are human testes located outside the abdominal cavity? Name the pouch in which they are present.

2. List the different parts of the human oviduct through which the ovum travels till it meets the sperm for fertilisation.

3. (i) Draw a labelled diagrammatic view of human male reproductive system,

(ii) Differentiate between vas deferens and vasa efferentia.

4. Explain the role of pituitary and sex hormones in the process of spermatogenesis.

OR

Spermatogenesis in human males is a hormone-regulated process. Justify.

OR

Explain the hormonal regulation of the process of spermatogenesis in humans.

5. Draw and label the parts of the head region only of a human sperm.

6. Differentiate between menarche and menopause.

7. Explain the events in a normal woman during her menstrual cycle on the following days

(i) Ovarian event from 13-15 days.

(ii) Ovarian hormones level from 16-23 days.

(iii) Uterine events from 24-29 days.

8. (i) Explain the menstrual cycle in human females.

(ii) How can the scientific understanding of the menstrual cycle of human females help as a contraceptive measure?

9. (i) Explain the process of spermatogenesis in humans.

(ii) Draw a human sperm and label the acrosome and middle piece. Mention their functions.

10. Briefly explain the events of fertilisation and implantation in an adult human female.

(ii) Comment on the role of the placenta as an endocrine gland.

Or

Explain the process of fertilisation and implantation in humans.

11. Charu got pregnant with her first baby. She asked her husband (who is a doctor) to know about the changes which would take place in her body during pregnancy.

(i) What is implantation?

(ii) What values does the Charu's husband show?

Chapter 4 Reproductive Health

1. Our government has intentionally imposed strict conditions for MTP in our country. Justify giving a reason.

2. Describe the lactational amenorrhea method of birth control.

3. At the time of Independence, the population of India was 350 million, which exploded to over 1 billion by May 2000.

List any two reasons for this rise in population and any two steps taken by the government to check this population explosion.

4. If implementation of better techniques and new strategies are required to provide more efficient care and assistance to people, then why is there a statutory ban on amniocentesis? Write the use of this technique and give a reason to justify the ban.

5. An infertile couple is advised to adopt a test tube baby programme. Describe two principal procedures adopted for such technologies. **(Delhi 2015)**

OR

Explain the Zygote Intra Fallopian Transfer Technique (ZIFT). How is the Intra Uterine Transfer (IUT) Technique different from it?

6. Suggest and explain any three Assisted Reproductive Technologies (ART) to an infertile couple.

7. A large number of married couples in the world are childless. It is shocking to know that in India, the female partner is often blamed for the couple being childless.

(i) Why in your opinion the female partner is often blamed for such situations in India? Mention any two values that you as a biology student can promote to check this social evil.

(ii) State any two reasons responsible for the cause of infertility.

(iii) Suggest a technique that can help the couple to have a child where the problem is with the male partner.

Chapter 5 Principles of Inheritance and Variation

1. Name the pattern of inheritance where the F1phenotype

(i) resembles only one of the two parents.

(ii) does not resemble either of the two parents and is in between the two.

2. Mention any two contrasting traits with respect to seeds in pea plants that were studied by Mendel.

3. A garden pea plant (A) produced an inflated yellow pod and another plant (B) of the same species produced constricted green pods. Identify the dominant traits.

4. Why did TH Morgan select Drosophila melanogaster to study sex-linked genes for his lab experiments?

OR

Write the scientific name of the fruit fly. Why did Morgan prefer to work with fruit flies for his experiments? State any three reasons.

5. Linkage and crossing over of genes are alternatives to each other. Justify with the help of an example.

6. In snapdragon, a cross between true-breeding red-flowered (RR) plants and true-breeding white-flowered (rr) plants showed a progeny of plants with all pink flowers.

(i) The appearance of pink flowers is not known as blending. Why?

(ii) What is this phenomenon known as?

7. In a dihybrid cross, white-eyed, yellow-bodied female Drosophila crossed with red-eyed, brown-bodied male Drosophila produced in F2-generation 1.3% recombinants and 98.7% progeny with parental type combinations. This observation of Morgan deviated from the Mendelian F2-phenotypic dihybrid ratio. Explain, giving reasons for Morgan's observation.

8. (i) Explain linkage and recombination as put forth by TH Morgan based on his observations with the Drosophila melanogaster crossing experiment.

(ii) Write the basis on which Alfred Sturtevant explained gene mapping.

9. Explain with the help of a suitable example, the inheritance of a trait where two different dominant alleles of a trait express themselves simultaneously in the progeny. Name this kind of inheritance pattern.

10. State and explain the 'law of independent assortment' in a typical Mendelian dihybrid cross. **(Delhi 2017)**

OR

Using Punnett square show the F2 result of a dihybrid cross where the pure breed parents have contrasting traits with reference to seed shape and seed colour in Pisum sativum. Give the phenotypic ratio.

Chapter 6 Molecular Basis of Inheritance

1. Why is RNA more reactive in comparison to DNA?

2. Name the enzyme and state its property that is responsible for continuous and discontinuous replication of the two strands of a DNA molecule.

3. What is the origin of replication in a chromosome? State its function.

4. Describe the structure of a nucleosome. **(Delhi 2017)**

OR

Draw a labelled diagram of a nucleosome. Where is it found in a cell?

(Foreign 2014; All India 2012)

OR

How do histones acquire a positive charge?

5. Differentiate between exons and introns.

6. (i) Describe the structure and function of a tRNA molecule. Why is it referred to as an adapter molecule?

(ii) Explain the process of splicing of hnRNA in a eukaryotic cell. **(All India 2017)**

7. Sketch a schematic diagram of the lac operon in a switched-on position. How is the operon switched off? Explain.

8. (i) Why is tRNA called an adapter?

(ii) Draw and label a secondary structure of tRNA. What does the actual structure of tRNA look like?

9. Given below are the sequences of nucleoside in a particular mRNA and amino acids coded by it.

UUUAU GUUC GAGUUAGU GUAA

Phe – Met – Phe – Glu – Leu – Val

Write the properties of genetic codes that can be and cannot be correlated from the above-given data.

Chapter 7 Evolution

1. Name the type of evolution that has resulted in the development of structures like the wings of a butterfly and birds. What are such structures called?

2. During the biology lecture on theories of evolution, Mrs. Sharma was teaching the topic of Lamarckism. Kapil was in a confused state after the lecture, so he went to his teacher and asked about his doubts. He asked if characters can pass on to the next generation, and why he does not have the tattoo that his mother had on her arm.

The teacher smiled and told him the actual facts about it. She also explained to him the reasons for the rejection of Lamarck's theory.

(i) Name the book in which Lamarckism was explained.

(ii) What are the three postulates of this theory?

(iii) What are the values shown by the teacher?

3. Explain the salient features of Hugo de Vries's theory of mutation. How is Darwin's theory of natural selection different from it? Explain.

4. (i) How does the Hardy-Weinberg equation explain genetic equilibrium?

(ii) Describe how this equilibrium gets disturbed which may lead to the founder effect.

OR

(i) Describe the Hardy-Weinberg principle

(ii) List any four factors, which affect genetic equilibrium.

(iii) Describe the founder effect.

5. Branching descent and natural selection are the two key concepts of the Darwinian theory of evolution. Explain each concept with the help of a suitable example.

6. Describe the three different ways by which natural selection can affect the frequency of a heritable trait in a population. **(Foreign 2014)**

OR

Explain the three ways in which natural selection operates on different traits in nature.

7. Explain the phenomenon of evolution by natural selection as supported by the variations observed in white-winged and dark-winged moth populations in England between 1850 and 1920.

8. Name the first human-like hominid. Mention his food habits and brain capacity.

9. Differentiate between homology and analogy. Give one example of each.

OR

Differentiate between divergent and convergent evolution. Give one example of each.

10. Explain with the help of an example the type of evolution homology is based on.

OR

Divergent evolution leads to homologous structures. Explain with the help of an example.

Chapter 8: Human Health and Diseases

1. You have attended a birthday party hosted by one of your classmates. You found some guests at the party sitting in a corner making a lot of noise and consuming 'something'. After a while one of the boys from the group started screaming, behaving abnormally and sweating profusely.

(i) Would you inform your parents/school authorities? Yes/No? Give a reason in support of your answer.

(ii) Prepare a note to be circulated amongst the schoolmates about the sources and dangers of any two drugs.

(iii) Write any two ways that you will suggest to your school principal, to promote awareness amongst the youth against the use of these drugs.

2. It is commonly observed that parents feel embarrassed to discuss freely with their adolescent children about sexuality and reproduction. The result of this parental inhibition is that the children go astray sometimes.

(i) Explain the reasons that you feel are behind such embarrassment amongst some parents to freely discuss such issues with their growing children.

(ii) By taking one example of a local plant and animal, how would you help these parents overcome such inhibitions about reproduction and sexuality?

3. 'Prevention is better than cure' is an apt slogan to safeguard adolescents from drug abuse. List any 6 steps that could be taken in this regard.

4. A person in your colony has recently been diagnosed with AIDS. People/residents in the colony want him to leave the colony for the fear of the spread of AIDS.

(i) Write your view on the situation, giving reasons.

(ii) List the possible preventive measures that you would suggest to the residents of your locality in a meeting organised by you so that they understand the situation.

(iii) Write the symptoms and the causative agent of AIDS.

5. (i) Name and explain giving reason, the type of immunity provided to the newborn by the colostrum and vaccinations.

(ii) Name the type of antibody

(a) present in colostrum.

(b) produced in response to allergens in the human body.

6. (i) HIV and Hepatitis B are STDs. Mention the two other ways by which they can be transmitted to a healthy person.

(ii) Why is early detection of STDs essential? What can it lead to otherwise? Explain.

7. Name the parasite that causes filariasis in humans. Mention its two diagnostic symptoms. How is this transmitted to others?

8. Differentiate between active and passive immunity.

9. State the function of interferons.

Chapter 10 Microbes in Human Welfare

1. (i) How do organic farmers control pests? Give two examples.

(ii) State the difference in their approach from that of conventional pest control methods.

2. How does the activity of each one of the following help in organic farming?

(i) Mycorrhiza

(ii) Cyanobacteria

(iii) Rhizobium

3. Draw a labelled sketch of a typical biogas plant.

4. State one reason for adding blue-green algae to the agricultural soil.

5. How can sewage be used to generate biogas? Explain.

Chapter 11 Biotechnology- Principles and Processes

1. Why is it not possible for an alien DNA to become part of a chromosome anywhere along its length and replicate normally?

2. Why is making cells competent essential for biotechnology experiments? List any two ways by which this can be achieved.

3. Industrial production of biologically important recombinant products utilises bioreactors. Sohan was aware of this information but was curious to know about the process involved. He asked his teacher about this, who explained the process in detail to him.

(i) What are bioreactors? Name the most common type of bioreactor utilised by industries.

(ii) What is the sequence of events after the completion of the biosynthetic phase in the bioreactor?

4. Write the steps you would suggest to be undertaken to obtain a foreign-gene product.

5. How are the following used in biotechnology?

(i) Plasmid DNA

(ii) Recognition sequence

(iii) Gel electrophoresis

6. (i) Explain the significance of palindromic nucleotide sequences in the formation of recombinant DNA.

(ii) Write the use of restriction endonuclease in the above process.

7. Explain the role(s) of the following in biotechnology

(i) Restriction endonuclease

(ii) Gel-electrophoresis

(iii) Selectable markers in pBR322

8. List the key tools used in recombinant DNA technology.

9. Explain the mode of action of Eco RI.

OR

What is Eco RI? How does Eco RI differ from an exonuclease?

OR

How does a restriction nuclease function? Explain.

10. Why do DNA fragments move towards the anode during gel electrophoresis?

Chapter 12 Biotechnology and its Applications

1. Biopiracy should be prevented. State why and how?

2. Mention any four benefits derived from transgenic animals.

3. (i) Name the source from which insulin was extracted earlier. Why is this insulin no longer in use by diabetic people?

(ii) Explain the process of synthesis of insulin by Eli Lilly company. Name the technique used by the company.

(iii) How is the insulin produced by the human body different from the insulin produced by the above-mentioned company?

4. Name the host plant and its part that Meloidogyne incognita infects. Explain the role of Agrobacterium in the production of dsRNA in the host plant.

5. Explain the various steps involved in the production of artificial insulin.

OR

Recombinant DNA technology is of great importance in the field of medicine. With the help of a flow chart, show how this technology has been used in preparing genetically engineered human insulins.

6. How does 'RNA interference' take place in eukaryotes? Mention its importance. **(Delhi 2012C)**

OR

Explain the process of RNA interference.

7. Write the functions of

(i) cry IAc gene

(ii) RNA interference (RNAi)

Chapter 13: Organisms and Populations

1. (i) Explain with the help of a graph the population growth curve, when resources are (A) limiting (B) not limiting.

(ii) Nature has a carrying capacity for a species. Explain.

2. What is the association between the bumble bee and its favourite orchid Ophrysl How would the extinction or change of one affect the other?

3. During a school trip to 'Rohtang Pass', one of your classmate suddenly developed 'altitude sickness'. But, he/she recovered after some time.

(i) Mention one symptom to diagnose the sickness.

(ii) What caused the sickness?

(iii) How could she/he recover by her/himself after some time?

4. Explain with the help of suitable examples the three different ways by which organisms overcome their stressful conditions lasting for short duration.

OR

Explain by taking three different examples how do certain organisms pull through the adverse conditions when unable to migrate under stressful period.

5. Following are the responses of different animals to various abiotic factors. Describe each one with the help of an example.

(a) Regulate

(b) Conform

(c) Migrate

(d) Suspend

(ii) If 8 individuals in a population of 80 butterflies die in a week, calculate the death of the population of butterflies during that period.

6. (i) Explain with the help of a graph the population growth curve, when resources are (A) limiting (B) not limiting.

(ii) Nature has a carrying capacity for a species. Explain

Chapter 14 Ecosystem

1. How does a detritivore differ from a decomposer? Explain with an example of each.

2. Write the equation that helps in deriving the net primary productivity of an ecosystem. **(Delhi 2013)**

OR

How is the net primary productivity of an ecosystem derived?

3. Write the relationship between productivity, gross primary productivity, net primary productivity and secondary productivity.

4. (i) Healthy ecosystems are the base of a wide range of (ecosystem) services. Justify.

(ii) Explain the differences and the similarities between hydrarch and xerarch successions of plants.

5. Explain the carbon cycle with the help of a simplified model.

6. How does the phosphorus cycle differ from the carbon cycle?

Chapter 15 Biodiversity and Conservation

1. (i) Why should we conserve biodiversity? How can we do it?

(ii) Explain the importance of biodiversity hotspots and sacred groves.

2. List six advantages of 'ex situ' approach to conservation of biodiversity.

3. Write the importance of cryopreservation in conservation of biodiversity.

4. Write the importance of species diversity to the ecosystem. Support your answer with the finding of Tilman.

5. Name the type of biodiversity represented by the following

(i) 1000 varieties of mangoes in India.

(ii) Variations in terms of potency and concentration of reserpine in Rauwolfia vomitoria growing in different regions of "Himalayas.